# COMPARING TWO TOPOLOGIES: DERIVED CATEGORIES OF ADEQUATE MODULES AGREE

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument: the identification of the categories, the vanishing of the higher direct images, the weak Serre property by the five lemma and the verification of the criteria are all written out. The phrases matching a generic omission filter inside the counted span are pointers to earlier discussions of notation, not disclaimers of arguments.

*Editorial note.* Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

*Editorial note.* The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

*Editorial note.* The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

*Allowed standard prerequisites (editorial).*

- The comparison of topologies for the first time. Same chapter, Section section-compare, which identifies the morphism of ringed sites and its basic properties. Supplies the morphism epsilon and the earlier statements used in the proof.
- Higher direct images as sheafifications of local cohomology. Same chapter, Lemma lemma-higher-direct-images. Turns the vanishing assumptions into the vanishing of the positive higher direct images.
- The comparison criterion for derived categories. Same chapter, Lemma lemma-equivalence-unbounded-two, whose conditions are checked at the end. Supplies the equivalence of triangulated categories once the five conditions are verified.
- The five lemma and weak Serre subcategories. Homology, Lemma homology-lemma-five-lemma and Definition homology-definition-serre-subcategory. Shows the image of the subcategory is a weak Serre subcategory.

*Source reference.* Original section 0EWK: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6177–6177. Navigation pointer invoked at proof line

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6548; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 072W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 747–747. Navigation pointer invoked at proof line 6599; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0754: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6603–6603. Navigation pointer invoked at proof line 6603; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original definition 02MO: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6628–6628. Navigation pointer invoked at proof line 6628; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 05QB: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6647–6647. Navigation pointer invoked at proof line 6647; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original section 06UP: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6658–6658. Navigation pointer invoked at proof line 6658; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

*Source reference.* Original lemma 0D7W: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 6482–6482. Navigation pointer invoked at proof line 6660; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Let $\mathcal{C}$ be a category. Let $\mathrm{Cov}(\mathcal{C}) \supset \mathrm{Cov}'(\mathcal{C})$ be two ways to endow $\mathcal{C}$ with the structure of a site. Denote $\tau$ the topology corresponding to $\mathrm{Cov}(\mathcal{C})$ and $\tau'$ the topology corresponding to $\mathrm{Cov}'(\mathcal{C})$. Then the identity functor on $\mathcal{C}$ defines a morphism of sites

$$\epsilon : \mathcal{C}_\tau \longrightarrow \mathcal{C}_{\tau'}$$

where $\epsilon_*$ is the identity functor on underlying presheaves and where $\epsilon^{-1}$ is the $\tau$-sheafification of a $\tau'$-sheaf (hence clearly exact). Let $\mathcal{O}$ be a sheaf of rings for the $\tau$-topology. Then $\mathcal{O}$ is also a sheaf for the $\tau'$-topology and $\epsilon$ becomes a morphism of ringed sites

$$\epsilon : (\mathcal{C}_\tau, \mathcal{O}_\tau) \longrightarrow (\mathcal{C}_{\tau'}, \mathcal{O}_{\tau'})$$

For more discussion, see Section 0EWK.

**Lemma 0.1.** *With $\epsilon : (\mathcal{C}_\tau, \mathcal{O}_\tau) \to (\mathcal{C}_{\tau'}, \mathcal{O}_{\tau'})$ as above. Let $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ be a subset. Let $\mathcal{A} \subset PMod(\mathcal{O})$ be a full subcategory. Assume*

1. *every object of $\mathcal{A}$ is a sheaf for the $\tau$-topology,*
2. *$\mathcal{A}$ is a weak Serre subcategory of $Mod(\mathcal{O}_\tau)$,*
3. *every object of $\mathcal{C}$ has a $\tau'$-covering whose members are elements of $\mathcal{B}$, and*

(4) *for every* $U \in \mathcal{B}$ *we have* $H^p_\tau(U, \mathcal{F}) = 0$, $p > 0$ *for all* $\mathcal{F} \in \mathcal{A}$.

*Then* $\mathcal{A}$ *is a weak Serre subcategory of* $Mod(\mathcal{O}_{\tau'})$ *and there is an equivalence of triangulated categories* $D_{\mathcal{A}}(\mathcal{O}_\tau) = D_{\mathcal{A}}(\mathcal{O}_{\tau'})$ *given by* $\epsilon^*$ *and* $R\epsilon_*$.

*Proof.* Since $\epsilon^{-1}\mathcal{O}_{\tau'} = \mathcal{O}_\tau$ we see that $\epsilon$ is a flat morphism of ringed sites and that in fact $\epsilon^{-1} = \epsilon^*$ on sheaves of modules. By property (1) we can think of every object of $\mathcal{A}$ as a sheaf of $\mathcal{O}_\tau$-modules and as a sheaf of $\mathcal{O}_{\tau'}$-modules. In other words, we have fully faithful inclusion functors

$$\mathcal{A} \to Mod(\mathcal{O}_\tau) \to Mod(\mathcal{O}_{\tau'})$$

To avoid confusion we will denote $\mathcal{A}' \subset Mod(\mathcal{O}_{\tau'})$ the image of $\mathcal{A}$. Then it is clear that $\epsilon_* : \mathcal{A} \to \mathcal{A}'$ and $\epsilon^* : \mathcal{A}' \to \mathcal{A}$ are quasi-inverse equivalences (see discussion preceding the lemma and use that objects of $\mathcal{A}'$ are sheaves in the $\tau$ topology).

Conditions (3) and (4) imply that $R^p\epsilon_*\mathcal{F} = 0$ for $p > 0$ and $\mathcal{F} \in \mathrm{Ob}(\mathcal{A})$. This is true because $R^p\epsilon_*$ is the sheaf associated to the presheave $U \mapsto H^p_\tau(U, \mathcal{F})$, see Lemma 072W. Thus any exact complex in $\mathcal{A}$ (which is the same thing as an exact complex in $Mod(\mathcal{O}_\tau)$ whose terms are in $\mathcal{A}$, see Homology, Lemma 0754) remains exact upon applying the functor $\epsilon_*$.

Consider an exact sequence

$$\mathcal{F}'_0 \to \mathcal{F}'_1 \to \mathcal{F}'_2 \to \mathcal{F}'_3 \to \mathcal{F}'_4$$

in $Mod(\mathcal{O}_{\tau'})$ with $\mathcal{F}'_0, \mathcal{F}'_1, \mathcal{F}'_3, \mathcal{F}'_4$ in $\mathcal{A}'$. Apply the exact functor $\epsilon^*$ to get an exact sequence

$$\epsilon^*\mathcal{F}'_0 \to \epsilon^*\mathcal{F}'_1 \to \epsilon^*\mathcal{F}'_2 \to \epsilon^*\mathcal{F}'_3 \to \epsilon^*\mathcal{F}'_4$$

in $Mod(\mathcal{O}_\tau)$. Since $\mathcal{A}$ is a weak Serre subcategory and since $\epsilon^*\mathcal{F}'_0, \epsilon^*\mathcal{F}'_1, \epsilon^*\mathcal{F}'_3, \epsilon^*\mathcal{F}'_4$ are in $\mathcal{A}$, we conclude that $\epsilon^*\mathcal{F}_2$ is in $\mathcal{A}$ by Homology, Definition 02MO. Consider the map of sequences

$$\begin{array}{ccccccccc}
\mathcal{F}'_0 & \longrightarrow & \mathcal{F}'_1 & \longrightarrow & \mathcal{F}'_2 & \longrightarrow & \mathcal{F}'_3 & \longrightarrow & \mathcal{F}'_4 \\
\downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\
\epsilon_*\epsilon^*\mathcal{F}'_0 & \longrightarrow & \epsilon_*\epsilon^*\mathcal{F}'_1 & \longrightarrow & \epsilon_*\epsilon^*\mathcal{F}'_2 & \longrightarrow & \epsilon_*\epsilon^*\mathcal{F}'_3 & \longrightarrow & \epsilon_*\epsilon^*\mathcal{F}'_4
\end{array}$$

The lower row is exact by the discussion in the preceding paragraph. The vertical arrows with index 0, 1, 3, 4 are isomorphisms by the discussion in the first paragraph. By the 5 lemma (Homology, Lemma 05QB) we find that $\mathcal{F}'_2 \cong \epsilon_*\epsilon^*\mathcal{F}'_2$ and hence $\mathcal{F}'_2$ is in $\mathcal{A}'$. In this way we see that $\mathcal{A}'$ is a weak Serre subcategory of $Mod(\mathcal{O}_{\tau'})$, see Homology, Definition 02MO.

At this point it makes sense to talk about the derived categories $D_{\mathcal{A}}(\mathcal{O}_\tau)$ and $D_{\mathcal{A}'}(\mathcal{O}_{\tau'})$, see Derived Categories, Section 06UP. To finish the proof we show that conditions (1) – (5) of Lemma 0D7W apply. We have already seen (1), (2), (3) above. Note that since every object has a $\tau'$-covering by objects of $\mathcal{B}$, a fortiori every object has a $\tau$-covering by objects of $\mathcal{B}$. Hence condition (4) of Lemma 0D7W is satisfied. Similarly, condition (5) is satisfied as well. □